MB Note

1 Dinamical Systems

1.1 Models of Biological Systems

Logistic Growth Model (Population): Model: $\frac{dN}{dt} = rN(1-\frac{N}{K})$ N: population at time t; K: carrying capacity (depends on resources); r,K: > 0.

Sol: $N(t) = \frac{N_0 K e^{rt}}{K+N_0(e^{rt}-1)} \xrightarrow{\infty} K$ **Equilibrium points:** $N=0$ (unstable), $N=K$ (stable). **Quickest:** $N=\frac{K}{2}$ (<: 凹; >: 凸)

Delay Model: Model: $\frac{dN}{dt} = f(N(t), N(t-T))$ T: delay time > 0 周期解 | 与初始值无关 ps: To consider e.g. gestation period, maturation time etc.

[1]**Ext. of Logistic Model:** $\frac{dN}{dt} = rN(t)\left[1-\frac{N(t-T)}{K}\right]$ (用"ave. pop." 估计, 准用"Convolution Type") $\Rightarrow$ [2]**Dimensionless:** $\frac{dN^*}{dt^*} = N^*(t^*)\left[1-N^*(t^*-T^*)\right]$ $(N^*=\frac{N}{K}, t^*=rt, T^*=rT)$

[1,2] **Periodic sol:** stable limit cycle $N(t+t_p)=N(t)$, t_p: period. **Amplitude:** rT **Equilibrium Points:** $N=0|N^*=0$ (unstable); $N=K|N^*=1$ (Stable: $0<T^*=rT<\frac{\pi}{2}$; Unstable: $T^*>\frac{\pi}{2}$)

Continuous Population Model (EN): Model: $\frac{dN}{dt} = rN(1-\frac{N}{K}) - EN = f(N)$ r,K,E: >0; EN: harvesting rate/unit time; E: effort measure.

Equilibrium: $N=0$ (unstable); $N=N_h(E)=K(1-\frac{E}{r})>0$ (need $E<r$, Stable) **Yield:** $Y(E)=EN$ $\cdot$ $E>r$:die out; else: **Max Yield:** $Y_M=\frac{rK}{4}$ $(E=\frac{r}{2})$ **Steady State:** $N_{h|Y_M}=\frac{K}{2}$

Continuous Population Model (Y_0): Model: $\frac{dN}{dt} = rN(1-\frac{N}{K}) - Y_0 = f(N; r, K, Y_0)$ Y_0: constant yield rate/unit time.

Max Yield: $Y_M=\frac{rK}{4}$ **Equilibrium:** [1]$0<Y_0<Y_M$: Two positive points (unstable(小) and stable(大)). **Sensitivity:** 对于 fluctuation 此模型和 EN 模型都不稳定, 但此模型更敏感 sensitive.

Examples of $N_{t+1}=f(N_t)$: Verhulst: $N_{t+1}=rN_t(1-\frac{N_t}{K}); r, K>0$. Better: **Ricker curve:** $N_{t+1}=N_t e^{r(1-\frac{N_t}{K})}$

Imperfect Bifuraction: $\dot{x}=rx-x^3+h$ (codimension-2 bifurcation) If $r\le 0$, x^* unique. If $r>0$, $|h|>h_c$, x^* unique; $|h|<h_c$, x^* three. (两边 stable, 中 unstable). Bifuraction points: $h=\pm h_c=\pm\frac{2r}{3}\sqrt{\frac{r}{3}}$. **cusp point**: $(r,h)=(0,0)$

1.2 Summary - Stability and Bifuraction

	$\frac{dN}{dt}=f(N)$ **No Limit cycle periodic solutions** S: stable; U:unstable.				$N_{t+1}=N_tF(N_t)=f(N_t)$ **Eigenvalue:** $\lambda=f'(N^*)$			
Stability	Stable: $f'(N^*)<0$		Unstable: $f'(N^*)>0$		Stable: $\lvert\lambda\rvert<1$	Unstable: $\lvert\lambda\rvert>1$ (chaotic)	Periodic: $\lvert\lambda\rvert=1$ $\begin{cases}+: \text{tangent bifurcation}\\ -: \text{period-doubling bifurcation}\end{cases}$	
Linearise	$N(t)=N^*+n(t), \lvert n(t)\rvert\ll 1 \Rightarrow \frac{dn}{dt}=\frac{dN}{dt}=\begin{cases}\text{(general)} = f(N^*)+f'(N^*)n(t)\\ \text{(delay)= 直接带入 f 后近似}\end{cases}$				Let $N_t=N^*+n_t, \lvert n_t\rvert\ll 1 \Rightarrow N^*+n_{t+1}=\cdots=f(N^*+n_t)\approx f(N^*)+f'(N^*)n_t \Rightarrow n_{t+1}=\lambda n_t$			
Bifuraction \\\\ Perturbations at N^* | Periodic	Saddle-Note (Normal Form) $\dot{x}=r\pm x^2$ 0→S→S-S	Transcritical $\dot{x}=rx-x^2$ S-U→U-S	(super-)Pitchfork $\dot{x}=rx-x^3$ S→S-U-S	Sub-critical Pitchfork $\dot{x}=r+x^3$ U→U-S-U	$\lambda<-1$: ↑ oscillations If $f^p(N^*)=N^*$, N^* is p-periodic. **Sarkovskii's Theorem**: If a 3-periodic N^* exists, then $\forall p\ge 1$, p-periodic N^* exists.	$-1<\lambda<0$: ↓ oscillations | $\forall$ even p-periodic N^* with $\lambda=-1$ branches into $2p$-periodic	$0<\lambda<1$: ↓ Monotone	$\lambda>1$: ↑ Monotone **Species extinction:** $N_{\text{min}}<1$
Other	**Time-Independent**: At N^*: $\frac{dN}{dt}=0$ $N(t)=N(t-T)=N^*$ $f(N^*)=0$				**Max/Min:** [如果单峰 (unimodal)|2-periodic] $f'(N_m)=0$, $N_{\text{max}}=f(N_m)$; $N_{\text{min}}=f(N_{\text{max}})$			
Other	**Tangency Condition:** If $\dot{x}=f(x,r)$, 分岔点 (x^*,r_c): $\frac{\partial f}{\partial x}\vert_{(x^*,r_c)}=0$ $f(x^*,r_c)=0$ | 两点合并:**coalesce**.							

Bifuraction: Qualitative changes in dynamics are called *bifurcations*. (当平衡点性质 | 个数发生变化时) **Bifuraction Diagram:** y 轴: 平衡点; x 轴: 控制参数; 虚线 unstable, 实线 stable.

稳定性 Stability 判别 |For Continuous Model: Find perturbation: $n(t)$ and $\frac{dn}{dt}$ $\Rightarrow$ Assume $n(t)=e^{\lambda t}|ce^{\lambda t}$ $\rightarrow$ Find $\lambda=g(\lambda)$ $\Rightarrow$ Let $\lambda=u+iw\rightarrow u=g_1(u)$ and $w=g_2(w)$ $\Rightarrow$ $u<0$ asy.stable; $u>0$ unstable. **Find T_c:** (如要寻找参数 (e.g. T) 范围/临界值 使得平衡点稳定) Check $T=0$ stable. $\Rightarrow$ Using $u=0$ (i.e. $\lambda=iw$) to find T_c.

2 Interacting Populations

Lotka-Volterra Model (Predator-Prey): $\frac{dN}{dt}=N(a-bP)$; $\frac{dP}{dt}=P(cN-d)$ $\Rightarrow$ nondim: $\frac{du}{d\tau}=u(1-v)$; $\frac{dv}{d\tau}=\alpha v(u-1)$ **Equilibrium:** $(u^*,v^*)=(0,0),(1,1)$

Exact sol: $\alpha u+v-\ln(u^\alpha v)=H$, $H>H_{min}=\alpha+1$ constant. (closed phase) **Conservation Law:** $\frac{dH}{d\tau}=0$, here it satisfies. (By chain rule)

Show Level curves (constant of motion) closed: If $H(u,v)$ is level curve, Look $\begin{bmatrix}H_{uu} & H_{uv}\\ H_{vu} & H_{vv}\end{bmatrix}_{(u^*,v^*)}$ $\Rightarrow$ If $+/-$ definite (i.e. $\forall\lambda_i>0(<0)$)

$\Rightarrow$ It's local min/max at (u^*,v^*) & strictly convex/concave $\Rightarrow$ closed curves. Closed curves $\Rightarrow$ periodic sol.

Determinant Stability: For model: $\begin{bmatrix}u(t)\\ v(t)\end{bmatrix}=\begin{bmatrix}f(u,v)\\ g(u,v)\end{bmatrix}$ at equilibrium (u^*,v^*). Let $\begin{bmatrix}x\\ y\end{bmatrix}=\begin{bmatrix}u-u^*\\ v-v^*\end{bmatrix}$, then $\begin{bmatrix}x'\\ y'\end{bmatrix}=\begin{bmatrix}f_u & f_v\\ g_u & g_v\end{bmatrix}_{(u^*,v^*)}\begin{bmatrix}x\\ y\end{bmatrix}$ $\Rightarrow$ Find eigenvalues λ of J by $\det(J-\lambda I)=0$

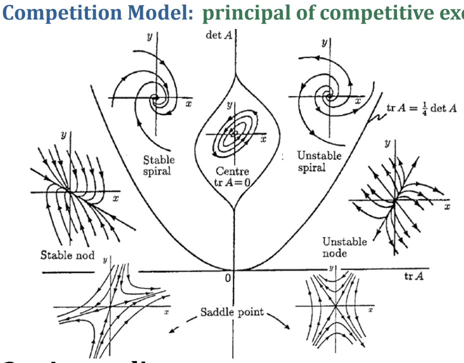
[1] $\det(J)>0, \text{tr}(J)<0 \Leftrightarrow \Re(\lambda)<0 \rightarrow$ stable node $_{Im(\lambda)=0}$; stable focus=spiral point $_{Im(\lambda)\neq 0}$; [2] $\det(J)>0, \text{tr}(J)>0 \Leftrightarrow \Re(\lambda)>0 \rightarrow$ unstable node $_{Im(\lambda)=0}$; unstable focus=spiral point $_{Im(\lambda)\neq 0}$

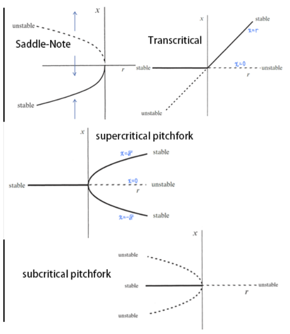
[3] $\det(J)<0 \Leftrightarrow Im(\lambda)=0, \lambda_1<0<\lambda_2 \rightarrow$ saddle point (unstable); [4] $\det(J)>0, \text{tr}(J)=0 \Leftrightarrow \lambda=\pm iw$ (pure imaginary) $\rightarrow$ center (neutrally stable)

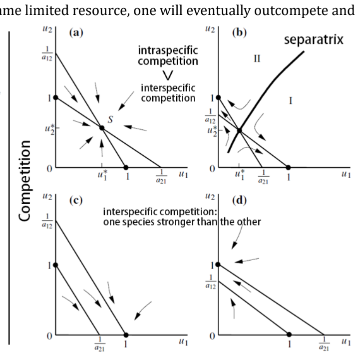
Null clines: $f(u,v)=0$ or $g(u,v)=0$ $\rightarrow$ 得 u,v 表达线条是 nullclines. **Trapping Region:** A closed region R 可以用直线/null clines 构成 s.t. 所有 边界的外法线方向 **n**, 在对应边界上的点, 满足 $\mathbf{n}\cdot\begin{bmatrix}f(u,v)\\ g(u,v)\end{bmatrix}<0$

Poincare-Bendixson Theorem: One unstable node/focus/spiral point + it's in *trapping region* $\Rightarrow$ A **stable limit cycle** exists. (Cannot be a saddle (since $\det(J)$ cannot be <0))

Competition Model: principal of competitive exclusion: If two species compete for the same limited resource, one will eventually outcompete and exclude the other. (具体四种情况看图)







Mutualism|Symbiosis: (共生)

Case I: 图 (a) coexist: competition not aggressive.

Case II: 两条射线 both species grow unboundedly.(all unstable)

Case II: 两条向外的线相交在一点 both species tend to a stable coexistence.

3 Appendix

$y'+p(x)y=q(x)$: **Integrating Factor:** $I(x)=e^{\int p(x)dx}$ $\Rightarrow$ $I(x)y=\int I(x)q(x)dx$ **Taylor Expansion:** $f(x,r)$ at (x_0,r_0): $f(x,r)=f(x_0,r_0)+(x-x_0)f_x|_{(x_0,r_0)}+(r-r_0)f_r|_{(x_0,r_0)}+\frac{1}{2}\cdots$

Binomial Series: $(1+x)^\alpha=1+\alpha x+\frac{\alpha(\alpha-1)}{2!}x^2+\cdots$ **Linerisation:** $f(x+\Delta x)=f(x)+f'(x)\Delta x+\cdots$ $e^{bx}=1+bx+\frac{(bx)^2}{2!}+\cdots$ $\ln(a+x)=\ln(a)+\frac{x}{a}-\frac{x^2}{2a^2}+\frac{x^3}{3a^3}-\cdots$

Hyperbolic Functions: $\sinh x=\frac{e^x-e^{-x}}{2}$ $\cosh x=\frac{e^x+e^{-x}}{2}$ $\tanh x=\frac{\sinh x}{\cosh x}$ $\sinh(x)=x+\frac{x^3}{3!}+\frac{x^5}{5!}+\cdots$ $\cosh(x)=1+\frac{x^2}{2!}+\frac{x^4}{4!}+\cdots$ $\tanh(x)=x-\frac{x^3}{3}+\frac{2x^5}{15}-\cdots$